

Turbulence-Resilient 120-Gbit/s Transmission Using SVD-Based Beamforming Towards Ground-to-LEO Optical Uplinks

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Abstract Using singular-value-decomposition (SVD)-based beamforming, we demonstrate turbulence-resilient 120-Gbit/s transmission over a 2.2-m link towards ground-to-LEO uplink scenarios. At 1550 nm, SVD provides >18-dB power enhancement over a Gaussian beam and BERs below 7% FEC threshold. SVD's tunability is verified across various turbulence realizations.

Introduction

High-throughput data transfer between Earth and space is crucial to meet the growing demand for satellite networks [1], [2]. Compared to RF links, free-space optical (FSO) communications offer higher bandwidth, greater directivity, and a lower probability of interception, but are much more sensitive to atmospheric effects [3]. Specifically, turbulence can cause scintillation, beam wander, and power redistribution, resulting in severe and varying power loss under a finite receiver (Rx) aperture and potentially inducing link outages [1]. Moreover, beam divergence over long distances further reduces the power collected by the finite Rx aperture [4]. Maximizing Rx power under such harsh channel conditions is a key challenge.

These degradations are asymmetric between uplinks and downlinks. Uplinks are particularly challenging because strong near-ground turbulence distorts the transmitted wavefront early in propagation [5]. Unlike in downlinks, these distortions subsequently propagate over hundreds of kilometers, accentuating the aforementioned effects at the satellite [1], [5], [6]. Indeed, reported

uplink capacity lags significantly behind downlink capacity [3], [7], [8].

Typically, FSO systems transmit data using a fundamental Gaussian mode and are subject to these impairments. However, tunable spatial beamforming may help maintain high collected power at a finite-aperture Rx, even under dynamic turbulence [9], [10]. Coherent beam combining (CBC)-based beamforming using multiple Gaussian beams has shown promising results in turbulence mitigation [11], [12]. Some CBC approaches iteratively adjust the phase at the transmitter (Tx) using feedback from the Tx or Rx power [12]. Although effective, such feedback-based optimization may need repeated measurements and generally neither reconstructs the channel matrix nor guarantees convergence to the globally optimal Tx field [13]. A laudable goal is therefore to find the optimal Tx field that maximizes the power collected by a finite-aperture Rx under a fixed Tx-power constraint, without iterative power-feedback optimization [13], [14], [15].

In this paper, we report singular-value decomposition (SVD)-based beamforming for turbulence-resilient 120-Gbit/s transmission towards ground-to-LEO optical uplink scenarios. For each channel realization, an orthogonal modal basis set, i.e., sub-aperture element basis, is used to reconstruct the spatial channel matrix. Its dominant right singular vector then defines a tunable, data-carrying Tx field that maximizes collected Rx power at fixed Tx power. A 2.2-m experimental link employs a near-Tx Kolmogorov phase plate ($r_0 = 0.4$ mm) to attempt to mimic near-ground turbulence. At 1550 nm, the SVD beam generated from 5×5 Tx elements provides >18-dB Rx-power gain over a Gaussian beam. The SVD beam supports a 120-Gbit/s 16-QAM transmission with BERs below the 7% FEC limit across multiple turbulence realizations.

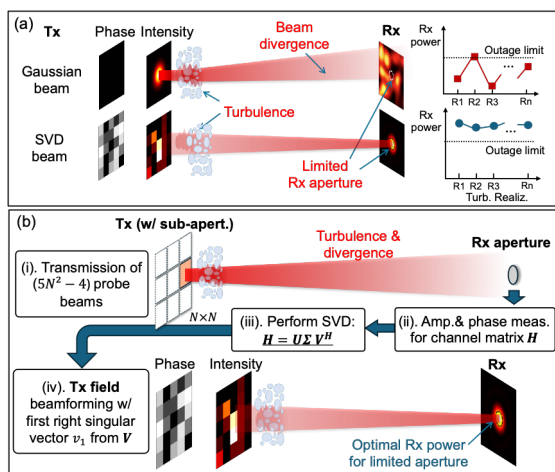


Fig. 1: (a) Comparison of Rx power of a Gaussian and SVD beam under harsh channel conditions with different turbulence realizations. (b) Concept of SVD-based Tx beamforming for an optimal Rx power.

Concept

FSO links typically use Gaussian beams. As

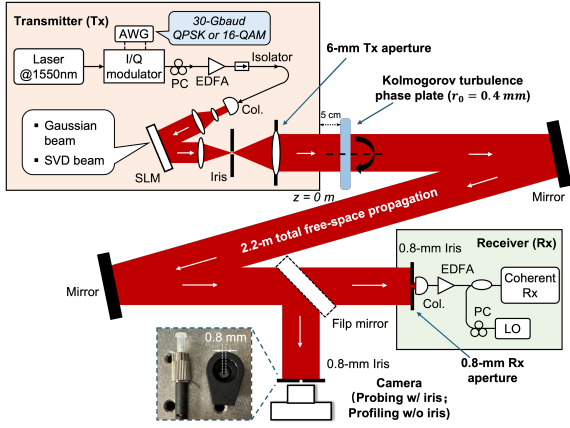


Fig. 2: Experimental setup. A flip mirror directs the light to the coherent Rx or the camera for channel probing, both with a 0.8-mm iris at a link length of 2.2m. Col.: collimator, PC: polarization controller, SLM: spatial light modulator.

shown in Fig. 1(a), challenging ground-to-LEO uplinks can involve three channel degradations: (i) near-ground turbulence distorts the Tx wavefront in the early stage of beam propagation [5]; (ii) beam divergence during uplink-type propagation expands and redistributes the distorted beam; and (iii) a finite, typically decimeter-scale satellite Rx aperture captures only a fraction of the distorted beam [16]. Altogether, these effects can potentially cause severe, time-varying power losses and link outages [5].

Figure 1(b) summarizes the SVD-beam synthesis. Consider a distributed Tx aperture that comprises $N \times N$ elements with fields E_i ($i = 1, \dots, N^2$). First, each element is illuminated individually to measure its amplitude response. Then, taking E_1 as the phase reference, each remaining element E_i ($i \neq 1$) is illuminated simultaneously with E_1 four times, with their relative phase set to $0, \pi/2, \pi$, and $3\pi/2$, respectively [17]. Thus, a total of $5N^2 - 4$ probes are transmitted and captured by the $M \times M$ camera pixels within the Rx aperture. Such four-step phase-shifting interferometry recovers the relative complex field response (amplitude and phase) of each Tx element [14], [17]. Stacking the response vectors yields $H = [h_1, \dots, h_{N^2}] \in \mathbb{C}^{M^2 \times N^2}$. For an arbitrary complex Tx weight vector $w \in \mathbb{C}^{N^2}$, the field

sampled within the Rx aperture is $y = Hw$. We decompose $H = U\Sigma V^H$, where U and V contain left and right singular vectors, and Σ contains singular values $\sigma_1 \geq \sigma_2 \geq \dots$. At fixed Tx power ($\|w\| = 1$), collected power is proportional to $\|Hw\|^2$ and is maximized by $w = v_1$: $\max \|Hw\|^2 = \sigma_1^2$, with $Hv_1 = \sigma_1 u_1$. Thus, if the channel remains quasi-static during probing and transmission, v_1 gives the unit-norm Tx weights that maximize finite-aperture Rx power.

We note that dynamic operation needs channel acquisition and feedback within the time that turbulence may change in a negligible way [12]. The number of probes needed for the proposed method could be reduced significantly using off-axis holography with a phase-stable reference at the Rx [18]. For an uplink spanning 100s of km, potential one-time probing incurs only a $\sim$ ms-scale round-trip delay [4]. Thus, the system may be able to update before turbulence changes in a non-negligible way. By applying a bidirectional link, a downlink beacon using optical or RF may further reduce the feedback time [19].

Experimental setup

Figure 2 shows the 2.2-m experimental setup. A 1550-nm data-carrying optical signal is directed to a spatial light modulator (SLM), which generates a Gaussian or SVD beam with the same Tx power. A near-Tx Kolmogorov phase plate with a laboratory Fried parameter of $r_0 = 0.4$ mm introduces varied wavefront distortions, attempting to mimic uplink near-ground turbulence [1], [2]. A 0.8-mm iris is used to represent a limited satellite Rx aperture. A flip mirror directs the propagated field either to a camera for channel probing or to the coherent Rx for data detection. For channel matrix acquisition, the SLM sequentially launches the single-element and phase-stepped pairwise probes. The camera records the corresponding aperture-limited intensity patterns, from which the complex response is recovered and assembled into the channel matrix [17]. The dominant right singular vector is then mapped to the SLM to form the data-carrying SVD beam. Both 30-Gbaud QPSK and 16-QAM data signals are tested.

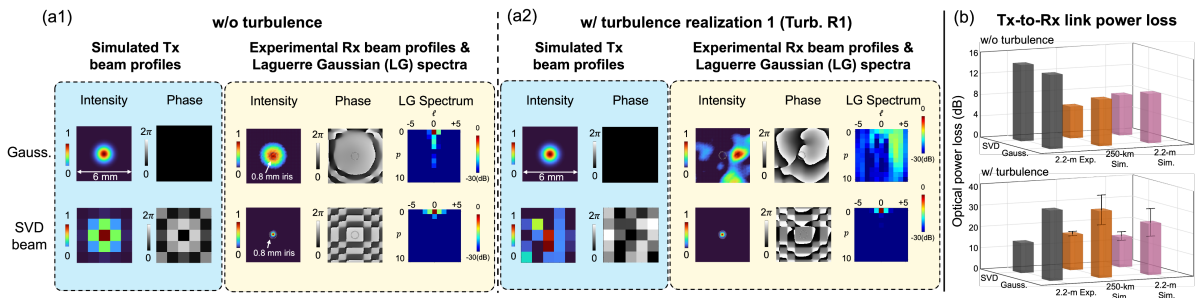


Fig. 3: (a) Measured Tx profiles (simulation), and Rx profiles and LG spectra (experiment) of Gaussian and SVD beam (a1) without and (a2) with turbulence. (b) Comparison of link power loss from Tx to Rx SMF in the 2.2-m experiment, a simulated 2.2-m link, and a 250-km ground-to-LEO link (b1) without and (b2) with turbulence. Simulations include 100 turbulence realizations.

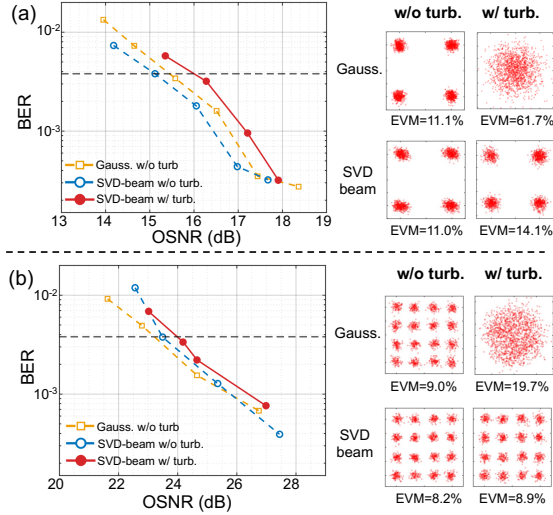


Fig. 4: Experimental BER performance, data constellations, and corresponding EVMs using a Gaussian and SVD beam for (a) 60-Gbit/s QPSK and (b) 120-Gbit/s 16-QAM signals.

Experimental results

Figure 3(a) shows the simulated Tx profiles and the measured Rx profiles and Laguerre-Gaussian (LG) spectra [20]. For fair comparison, we select a ~ 2 -mm waist that maximizes free-space Gaussian-beam power delivery between the 6-mm Tx and 0.8-mm Rx apertures [21]. The 5×5 SVD beam exhibits channel-dependent, discretized intensity and phase distributions across the controlled elements. Without turbulence (Fig. 3(a1)), the Gaussian beam remains symmetric, with its LG spectra concentrated near the azimuthal index $\ell = 0$, whereas the SVD output is tightly localized within the marked Rx aperture. With turbulence (Fig. 3(a2)), the Gaussian beam exhibits irregular intensity lobes and a distorted phase front, with LG spectra spanning radial (p) and azimuthal (ℓ) orders, consistent with its higher-order spatial components. The SVD output remains concentrated at the Rx aperture, with its LG spectrum dominated by low-order components. Although SVD optimization targets the Rx power [14], the resulting SVD beam also exhibits a relatively flat phase across the Rx aperture.

Figure 3(b) compares the measured and simulated link losses from Tx to a single-mode-fiber (SMF)-based Rx. A 250-km uplink simulation uses split-step propagation with 8 interleaved phase screens [22], whereas the 2.2-m one reproduces the single-screen experimental setup. The simulations exhibit similar beam-dependent trends without turbulence. With turbulence, however, the 250-km case shows ~ 5 dB higher loss, possibly due to stronger accumulated turbulence due to more phase screens [22]. Without turbulence, the measured losses exceed the simulated values, likely due to experimental alignment issues. With turbulence, the Gaussian beam

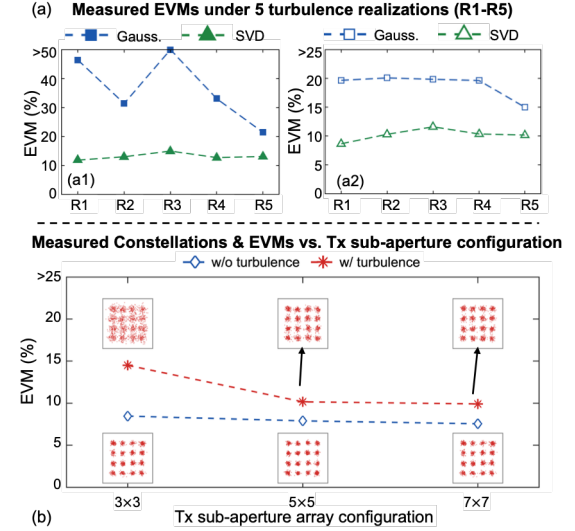


Fig. 5: (a) Experimentally measured EVMs of (a1) QPSK and (a2) 16-QAM signals under 5 turbulence realizations using different beams. (b) Measured constellations and EVMs under different Tx sub-aperture configurations.

exhibits ~ 33 -dB experimental link loss, whereas the SVD beam provides >18 -dB Rx power gain. Both simulations reproduce this relative advantage across 100 turbulence realizations.

Figure 4 shows the experimental link performance for 30-Gbaud QPSK and 16-QAM. As shown in Fig. 4(a), both beams produce well-resolved constellations with EVMs near 11% without turbulence. Under turbulence, the data carried by the Gaussian beam cannot be recovered, with EVM $>61\%$; while the SVD constellation is retained with an EVM of 14.1%. At a 7% FEC threshold, the SVD beam under turbulence induces a ~ 1 -dB OSNR penalty. Figure 4(b) shows the 16-QAM results at a gross rate of 120 Gbit/s. Similarly, a ~ 0.5 -dB penalty is observed for the SVD beam in turbulence, whereas the Gaussian beam data cannot be recovered due to a large power loss of ~ 33 dB (see Fig. 3(b)).

Figure 5(a1)-(a2) extends the comparison to 5 different turbulence realizations (R1-R5). For QPSK, SVD beams' EVMs remain $<14\%$, while most Gaussian EVMs remain $>30\%$, with only R5 at an EVM of 18.5%. For 16-QAM, the SVD EVMs are 9-12%, whereas the data cannot be recovered using a Gaussian beam (EVM of $\sim 20\%$) for R1-R4, with R5 showing an EVM of 15.0%.

Figure 5(b) compares the 16-QAM EVM using various Tx array configurations. Under turbulence R5, increasing the array from 3×3 to 5×5 lowers the EVM from 14.5% to 10.2%, while 7×7 provides a smaller additional enhancement to 9.8%. Without turbulence, EVM decreases from 8.5% to 7.6% as the number of arrays increases. This shows a practical trade-off between SVD performance and system complexity, with performance potentially converging once a certain number of array elements are used.

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